

SEMESTER V

UEI610: FUNDAMENTALS OF MICROPROCESSORS AND MICROCONTROLLERS

L	T	P	Cr
3	0	2	4.0

Course Objectives: The microprocessors and microcontrollers are an integral part of the embedded systems. The objective of this course is to expose the students to variety of processors and foster their skills to develop applications with microprocessor or microcontroller as the central element.

Introduction to Microprocessors and Microcontrollers: Evolution of microprocessor, Types of various architectures; Harvard and Von-Neumann, Reduced Instruction Set Computer (RISC) and Complex Instruction Set Computer (CISC) architectures, Advantages of RISC architecture, Architecture of 8085 and 8086 microprocessors.

ARM Processor Fundamentals and Architecture: Architecture, ARM General purpose Register set, Architectural inheritance, ARM programmer's model, Current program status register, Advanced Microcontroller Bus Architecture (AMBA), Special Function Registers, GPIO's, Timers and counters, Exception and Interrupt handling, Analog to digital converters, serial communication, direct memory access.

Introduction to pipeline, ARM 3, 5 and 7 stage pipeline, Pipeline hazards.

ARM programming in Assembly: Writing code in assembly, Instruction set coding, ARM addressing modes, Branch instructions: Branch and Branch with Link, Branch, Branch with Link and exchange, Data processing instructions, Load and store instructions, Load and store multiply instructions, Conditional execution, Programming applications, Thumb state, Thumb Programmers model, Thumb programming Applications.

Hardware and sensor interfacing: ARM Development Environment, Interfacing of ARM processors with LEDs, LCD, Relays, and sensors, Programming of processors using C/C++.

Laboratory works: Programming examples of ARM based processors, Programming and Application development around PIC/ARM, Interfacing with peripherals etc.

Course Learning Outcome (CLO):

After the successful completion of the course the students will be able to:

1. Analyze and explain the architecture and functionality of various types of processors, including their design components and operational mechanisms.
2. Develop and debug programs for ARM processors to perform computational tasks and control functions.
3. Create applications utilizing timers, GPIO, ADC modules, and communication interfaces to address real-world system requirements.
4. Design functional applications using ARM processors and integrate peripherals for embedded system solutions.

Text Books:

1. Sloss, Andrew, Dominic Symes, and Chris Wright. *ARM system developer's guide: designing and optimizing system software*. Elsevier, 2004.
2. Zhu, Yifeng. *Embedded Systems with ARM Cortex-M Microcontrollers in Assembly*. E-Man Press Llc, 2023.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments/ Tutorials/ Quizzes/ Lab Evaluations)	30